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To cite this article: Oscar G. Bukstein (2010) Clinical Practice Guidelines for Attention-Deficit/Hyperactivity Disorder: A Review, *Postgraduate Medicine*, 122:5, 69-77, DOI: [10.3810/pgm.2010.09.2203](https://doi.org/10.3810/pgm.2010.09.2203)

To link to this article: <https://doi.org/10.3810/pgm.2010.09.2203>



Published online: 13 Mar 2015.



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Clinical Practice Guidelines for Attention-Deficit/Hyperactivity Disorder: A Review

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Abstract: The prevalence and public health importance of attention-deficit/hyperactivity disorder (ADHD) as well as its impact on public health has led various groups to develop clinical practice guidelines (CPGs). Clinical practice guidelines are systematically developed statements designed to assist practitioner and patient decisions about appropriate health care for specific clinical circumstances. In this article, we review 9 CPGs, including their development process, the recommendation of specific clinical actions, and advocacy of recommendations based on limited evidence. Despite the differences between these CPGs for ADHD, they appear to be complementary and not inconsistent. The CPGs all recommend: a structured approach to diagnosis and treatment; attention to psychiatric comorbidity and other ADHD-related conflicts during an individual's lifespan; consideration of medication (usually stimulants) and psychosocial therapies; and follow-up and monitoring of patient response. A patient-centered approach is also recommended in the evaluation and treatment of ADHD, although it is less evidence based than recommended therapies.

Keywords: ADHD; clinical practice guidelines; treatment; stimulants

Introduction

Attention-deficit/hyperactivity disorder (ADHD) is a common neurobiological condition affecting 5% to 8% of school-aged children,¹⁻⁴ with symptoms persisting into adulthood.^{5,6} Psychiatrists, pediatricians, and other primary care physicians frequently are asked by parents and teachers to evaluate children, adolescent, or adults for ADHD. The prevalence and public health importance of ADHD has led various groups to develop clinical practice guidelines (CPGs), which are systematically developed statements designed to assist practitioner and patient decisions about appropriate health care for specific clinical circumstances.⁷ Clinical practice guidelines promote evidence-based health care and provide support and resources to doctors and health care professionals to help maintain the highest standards of medicine. The development of CPGs requires a process of review of the literature, rating of evidence, and review by experts and other stakeholders (ie, health care professionals, teachers, family members, etc.). In this article, we describe several of these CPGs and use the common elements to formulate a summary of common elements from the basic recommendations resulting from these CPGs. The Appraisal of Guidelines for Research & Evaluation (AGREE) collaboration is an international collaboration from researchers and policy makers who have developed a framework for assessing the quality of CPGs.⁸ Using an instrument, the ideal CPG can be graded in 6 domains (Table 1).

We selected (not systematically) 9 CPGs that had a range of methodologies for diagnosing children, adults, or both (Table 2). The CPGs that we discuss are:

1. American Academy of Pediatrics (AAP): Clinical Practice Guideline: Diagnosis and Evaluation of the Child With Attention-Deficit/Hyperactivity Disorder⁹

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- and Clinical Practice Guideline: Treatment of the School-Aged Child With Attention-Deficit/Hyperactivity Disorder¹⁰
2. American Academy of Child and Adolescent Psychiatry (AACAP): Practice Parameter for the Assessment and Treatment of Children and Adolescents With Attention-Deficit/Hyperactivity Disorder¹¹
 3. National Institute for Health and Clinical Excellence (NICE; United Kingdom): Attention-Deficit/Hyperactivity Disorder Diagnosis and Management of ADHD in Children, Young People, and Adults¹²
 4. Scottish Intercollegiate Guidelines Network: Management of Attention-Deficit and Hyperkinetic Disorders in Children and Young People¹³
 5. Royal Australasian College of Physicians: Australian Guidelines on Attention Deficit Hyperactivity Disorder (ADHD)¹⁴
 6. British Association for Psychopharmacology: Evidence-Based Guidelines for Management of Attention-Deficit/Hyperactivity Disorder in Adolescents in Transition to Adult Services and in Adults¹⁵
 7. European Society for Child and Adolescent Psychiatry (ESCAP): European Clinical Guidelines for Hyperkinetic Disorder – First Upgrade¹⁶
 8. The Texas Consensus Conference Panel on Medication Treatment of Childhood Attention-Deficit/Hyperactivity Disorder: Texas Children's Medication Algorithm Project (TCMAP) (revised in 2007)^{17,18}
 9. Magellan Health Services, Clinical Practice Guideline Task Force, Clinical Practice Guideline for Patients with Attention-Deficit/Hyperactivity Disorder¹⁹

Each of the first 6 guidelines were developed according to a defined process of research literature review, expert

Table 1. AGREE Domains for Assessing Clinical Practice Guidelines

- 1. Scope and purpose:** overall aim of the guideline, specific clinical questions, and the target patient population.
- 2. Stakeholder involvement:** the extent to which the guideline represents the views of its intended users.
- 3. Rigor of development:** the process used to gather and synthesize the evidence, the methods to formulate the recommendations, and to update them.
- 4. Clarity and presentation:** the language and format of the guideline.
- 5. Applicability:** the likely organizational, behavioral, and cost implications of applying the guideline.
- 6. Editorial independence:** the independence of the recommendations and acknowledgment of possible conflicts of interest from the guideline development group.

Abbreviation: AGREE, Appraisal of Guidelines for Research and Evaluation.

review, and development, whereas the TCMAP and ESCAP projects consist of a series of algorithms and broad subjective recommendations, developed as a result of a consensus conference of experts. Finally, the Magellan guidelines are a very broad description of treatment practices. The guidelines differ in their development process, their recommendations of specific clinical actions, and advocacy of recommendations based on limited evidence.

Description of the Clinical Practice Guidelines for ADHD

The AAP practice guidelines for ADHD are divided into 2 documents—one for the evaluation and diagnosis⁹ and another for the treatment of ADHD¹⁰—and are focused on school-aged children (6–12 years). These guidelines are currently being revised, with new documents to be published in 2011. The diagnosis/evaluation guidelines contain the following recommendations: 1) primary care physicians should initiate an evaluation for ADHD in children presenting with inattention, hyperactivity, impulsivity, academic underachievement, or behavior problems; 2) the diagnosis of ADHD requires that a child meet *Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition, Text Revision* (DSM-IV-TR) criteria; 3) the assessment of ADHD requires evidence directly obtained from parents or caregivers on the core symptoms of ADHD in various settings, the age of onset, duration of symptoms, and extent of functional impairment; 4) the assessment of ADHD requires evidence directly obtained from the classroom teacher (or other school professional) on the core symptoms of ADHD, duration of symptoms, extent of functional impairment, and associated conditions; 5) evaluation of the child with ADHD should include assessment for associated (coexisting) conditions; and 6) other diagnostic tests are not routinely indicated to establish the diagnosis of ADHD but may be used for the assessment of other coexisting conditions (eg, learning disabilities and mental retardation).

The treatment guidelines further recommend that primary care physicians should establish a treatment program that recognizes ADHD as a chronic condition. The treating physician, parents, and child, in collaboration with school personnel, should specify appropriate target outcomes to guide management. The physician should recommend stimulant medication and/or behavior therapy as appropriate to improve target outcomes in children with ADHD. When the selected management for a child with ADHD has not met target outcomes, physicians should evaluate the original diagnosis, appropriate treatments, adherence to the treatment

Table 2. Comparison of Clinical Practice Guidelines for ADHD

Guideline	Methodologies	Target	Presence of Strength of Recommendation	Adult	Child
AAP	Literature review Consensus recommendations	Pediatricians and PCPs dealing with children	No		X
AACAP	Literature review Expert consensus recommendations	Mental health professionals	Yes		X
NICE	Literature review Evidenced-based recommendations	Health care professionals	Yes	X	X
British Association for Psychopharmacology	Literature review Evidenced-based recommendations	Health care professionals	Yes	X	X
Scottish Intercollegiate Guidelines Network	Literature review Evidenced-based and expert consensus recommendations Algorithms	Health care professionals	Yes		X
Royal Australasian College of Physicians	Literature review Evidenced-based and expert consensus recommendations	Health care professionals	Yes	X	X
ESCAP	Consensus recommendations Algorithms	Health care professionals	No		X
TCMAP	Consensus recommendations Algorithms	Health care professionals	No		X
Magellan	Literature review Summary of other CGPs	Health care professionals Other health care organizations	No	X	X

Abbreviations: AACAP, American Academy of Child and Adolescent Psychiatry; AAP, American Academy of Pediatrics; ADHD, attention-deficit/hyperactivity disorder; CGP, clinical practice guidelines; ESCAP, European Society for Child and Adolescent Psychiatry; NICE, National Institute for Health and Clinical Excellence; PCP, primary care physician; TCMAP, Texas Children's Medication Algorithm Project.

plan, and presence of coexisting conditions. The physician should periodically provide a systematic follow-up for the child with ADHD. Target outcomes and adverse effects should be monitored, and information should be obtained from parents, teachers, and the child.

The AACAP Practice Parameter for ADHD¹¹ consists of 13 recommendations covering the need for screening, comprehensive assessment, and treatment. Screening for ADHD should be part of every mental health evaluation. Evaluation should consist of clinical interviews with the parent and patient, obtaining information about the patient's school or day-care functioning, evaluation for comorbid psychiatric disorders, and review of the medical, social, and family history. If the patient's medical history is unremarkable, laboratory or neurological testing are not indicated. Similarly, psychological and neuropsychological tests are not mandatory for the diagnosis of ADHD, but should be performed if the patient's history suggests low general cognitive ability or low achievement in language or mathematics

relative to the patient's intellectual ability. The physician should evaluate the patient with ADHD for the presence of comorbid psychiatric disorders.

Treatment should consist of a well-developed and comprehensive treatment plan, with the initial psychopharmacological treatment of ADHD consisting of a trial with an agent approved by the US Food and Drug Administration (FDA). Treatment of ADHD should continue as long as symptoms remain present and cause impairment. Patients should be periodically assessed to determine whether there is continued need for treatment or if symptoms have remitted. If none of the above agents result in satisfactory treatment of the patient with ADHD, the physician should carefully review the diagnosis and consider behavior therapy and/or the use of medications not approved by the FDA. The patient should be monitored for treatment-emergent side effects, including monitoring of height and weight throughout treatment. If a patient with ADHD has a robust response to psychopharmacological treatment and subsequently shows normative

functioning in academic, family, and social functioning, then psychopharmacological treatment of the ADHD alone is satisfactory. If a patient has a less than optimal response to medication, has a comorbid disorder, or experiences stressors in family life, then psychosocial treatment in conjunction with medication treatment is often beneficial. The AACAP published a pocket card condensing the ADHD parameter into diagnostic and treatment algorithms, key points, and a list of medications,²⁰ providing a more practical application of the AACAP ADHD CPG.

In the United Kingdom, the NICE guidelines on the diagnosis and management of ADHD¹² focus more on parental training and psychological interventions. For a diagnosis of ADHD, symptoms should meet the diagnostic criteria in DSM-IV-TR or International Classification of Diseases 10th Revision (ICD-10) (hyperkinetic disorder [HKD]) and be associated with at least moderate psychological, social, and/or educational or occupational impairment based on interview and/or direct observation in multiple settings. Symptoms should also be pervasive, occurring in ≥ 2 important settings, including social, familial, educational, and/or occupational settings.

Treatment and care should take into account individuals' needs and preferences, and (in children) those of their parents. All individuals with ADHD, including children, should have the opportunity to be involved in decisions about their care and treatment in partnership with their health care professionals. Medication treatment is not recommended for preschool children with suspected ADHD or for older children and adolescents with moderate ADHD. Instead, the parents of children and adolescents with ADHD should be offered a group parental training program with the option of group psychological treatment or social skills training for the child or young person, and individual psychological therapy for older adolescents. The NICE guidelines recommend medication treatment as the first-line intervention for children, adolescents, and young adults with severe ADHD as well as for adults with ADHD, and general practitioners should not initiate medication treatments for ADHD, although they may continue prescribing and monitoring such treatment once started by a qualified expert.

Medication treatment should always form part of a comprehensive care package that includes psychological and educational components. The NICE guidelines advocate the establishment of multidisciplinary specialist ADHD teams and/or clinics and the provision of behavioral interventions in the classroom by teachers with necessary training. When a decision has been made to treat children or adolescents/

young adults with ADHD with medications, methylphenidate should be considered for ADHD without significant comorbidity or for ADHD with comorbid conduct disorder. Atomoxetine should be considered for children/adolescents if tics, Tourette's syndrome, an anxiety disorder, stimulant misuse, or risk of stimulant diversion are present, or if methylphenidate has been ineffective at the maximum tolerated dose, or the child/adolescent is intolerant to low or moderate doses of methylphenidate. Medication treatment for adults with ADHD should always form part of a comprehensive treatment program that addresses psychological, behavioral, and educational or occupational needs. Following a decision to start medication treatment in adults with ADHD, methylphenidate should normally be tried first. Adults who do not seek a medication treatment should be able to access psychological help instead.

The Scottish Intercollegiate Guidelines Network¹³ for children and adolescents with ADHD and HKD combine evidenced-based recommendations with good practice points, which are based on the clinical experience of the guidelines development group. The principles of intervention are the primary focus of the good practice points.

Parents or caregivers of children with ADHD/HKD (and older children with ADHD/HKD) should be given information about ADHD/HKD and about possible interventions, including their potential risks and benefits, and health and education services communicate regularly to promote understanding of the difficulties of ADHD/HKD to ensure a consistent approach to the individual across settings and to monitor effectiveness of intervention(s). Additionally, if symptoms of ADHD are mild, physicians should consider behavioral approaches first. The remainder of the major recommendations are evidenced-based, with the strength of evidence ranging from A to D. Most of the medication recommendations are given an A rating: in school-aged children and adolescents/young adults with HKD (severe ADHD), medication is recommended. In school-aged children, adolescents, and young adults with ADHD/HKD, a combination of medication and behavioral treatments is recommended for comorbid symptoms of oppositional defiant disorder and/or aggressive behavior.

The recommendation for school-aged children, adolescents, and young adults with ADHD/HKD and comorbid generalized anxiety disorders receiving a combination of medication and behavioral treatments is given a B rating. The recommendation for behavioral parent training for preschool children with symptoms of ADHD/HKD is given a B rating. Assessment recommendations are given C and D ratings as

well as good practice points. Similar to previously discussed guidelines, the Royal Australasian College of Physicians Guidelines¹⁴ recommend that the diagnosis be based on the DSM-IV-TR criteria and that the diagnosis of ADHD should be made only after a comprehensive assessment. This includes medical, developmental, and psychosocial assessment, evidence of impairment in multiple settings, and information from multiple informants.

For the management of ADHD, individuals with ADHD and their families should be provided with information and education about ADHD and its impact, as well as the advantages and disadvantages of potential treatment strategies. Multimodal therapy is recommended for the treatment of ADHD in all age groups, including psychosocial management strategies, medication, and educational interventions. The individualized management plan should be developed in collaboration with the individuals with ADHD as well as his or her parents and teachers. Such a comprehensive plan should take into account the specific needs and expressed preferences of the patient, and the circumstances of his or her family and culture; the associated psychosocial problems, educational difficulties, and comorbid conditions; and the suitability of the plan for the individual and his or her family when considering factors such as affordability, accessibility, and acceptability. Physicians should be alert to the risk of depression or other psychiatric disorders in parents/caregivers of children or adolescents with ADHD, and should refer them for support and treatment.

The Australian guidelines state that not all individuals with ADHD will require pharmacological management. Medications should be used only when symptoms are pervasive across settings (eg, school and home), cause significant academic, social, or behavioral impairments, and after careful consideration of nonpharmacological approaches. Clearly defined goals should be identified prior to initiating a trial of medication treatment. Medication should not be used as first-line treatment for ADHD in preschool-aged children. Patients receiving treatment for ADHD should be reviewed regularly (at least bi-annually) to ensure that the management strategies remain appropriate and effective.

Other therapies, such as elimination and restriction diets, are not supported as a general treatment for individuals with ADHD, although a subset of children who are sensitive to certain foods or food additives may benefit from careful exclusion diets. Other alternative treatments (including fatty acid supplements, biofeedback, homeopathy, or sensory integration diets) are not currently supported as treatments for individuals with ADHD.

The British Association for Psychopharmacology guidelines¹⁵ were developed for adolescents in transition to adult services, and in adults. Each of the recommendations is graded according to the supporting evidence. Diagnosing ADHD in adults requires a specialist (ie, a psychiatrist), but should also involve primary care practitioners, who should be trained to be aware of the diagnosis. Assessment includes identifying symptoms (past and present), impairments in different contexts, the influence of changing demands throughout the patient's lifespan, the exclusion of other problems (including psychiatric disorders), and application of clinical examination, rating scales, and other tools as indicated. Diagnostic and assessment criteria should be based on checklists based on DSM-IV-TR and/or ICD-10 with adult symptoms as appropriate. Assessment needs to confirm impairment in different domains, including: 1) a history of childhood ADHD or suggestive symptoms (impairments/failure of attainment/demands and comorbidity), and 2) current evidence of symptoms leading to pervasive impairment in > 1 domain (given context demands and skills). Multiple informants need to be used where possible (at least one contemporary and one developmental) with consent, especially for younger patients. Neuropsychological tests based solely on executive function are likely to be of limited diagnostic value, although test batteries that assess multiple domains of neuropsychological performance may be useful to determine individual deficits and to suggest tailored management strategies. Treatment response cannot be used to make a diagnosis of ADHD. Physicians need to screen for autistic spectrum, developmental disorders, communication difficulties, learning disabilities, tics and Tourette's syndrome, anxiety and affective disorders, substance use disorders, and other conditions (eg, epilepsy, sleep disorders, sensory problems), but symptoms cannot be counted twice for ADHD and comorbid disorders. Further specialist input or advice may be needed to confirm comorbidity (eg, learning disability teams), because comorbidity can have a greater impact on functioning than ADHD.

In the United Kingdom, prescribing medication for adult patients with ADHD is off-label because no agent is licensed there for this indication. However, atomoxetine is licensed for use in adults, but only when ADHD treatment was initiated in childhood. Proven medication treatments in children include psychostimulants and atomoxetine as first-line treatments, imipramine and bupropion as second-line treatments, and clonidine and guanfacine as possible adjunctive treatment. Although adult patients with ADHD are most likely to present to primary care, medication treatment is

best initiated and optimized by secondary/specialist services. Medication treatment needs to be chosen and adapted to best fit the individual, including the patient's preferences and concerns. Medication treatment requires regular (preferably structured) monitoring and review (eg, for dose adjustment) at least (for uncomplicated cases) twice a year or annually. These regular reviews include monitoring of ADHD and other symptoms, global and specific functioning, adverse effects, concordance of effects (eg, between patient, doctor, informants), psychiatric side effects, cardiovascular effects, compliance, and tolerance (daily and long term). Medication treatment should not be initiated if the diagnosis is uncertain or benefit is unlikely. Abuse potential is related to medication action and formulation. Slow-release preparations of these agents or atomoxetine are to be preferred for patients with a history of, or who are at risk of, medication misuse. While abuse by patients appears to be low, diversion can occur with stimulants for performance enhancement or weight loss.

Psychotherapeutic support to the individual, family, and others is helpful to inform on the condition and prognosis, to prevent negative effects on self-esteem or unrealistic expectations of treatment, to adapt positive and negative coping strategies, and to give perspective to individual neurodevelopmental history. Structured adapted psychotherapies may be useful to build confidence, develop executive skills, address anxiety and depression, and improve functioning, while group therapy may help to manage social isolation. Evidence from the literature on children with ADHD suggests that psychotherapies are beneficial for comorbid anxiety and for functional outcomes beyond the core symptoms, when added to medication treatment. Involvement of educational/occupational psychologists and other relevant personnel for environmental restructuring can maximize functioning at college or work. The guideline provides recommendations for individuals with specific problems such as learning disabilities and juvenile delinquency, as well as psychiatric comorbidity, including anxiety, depression, bipolar disorder, autistic spectrum disorder, conduct and oppositional defiant disorder, and substance use disorders.

The ESCAP revised guidelines¹⁶ were established from discussions between a group of child psychiatrists and psychologists from several European countries at the European Network for Hyperkinetic Disorders, in addition to an iterative critique of each clinical analysis. The participants were guided by evidence-based information, and the guidelines were based on evaluation (rather than meta-analysis) of the scientific evidence. When reliable information was lacking, the group

provided a clinical consensus. The document consists of a chapter-like review of assessment and treatment options and a subjective, general discussion of the supporting evidence for a general scheme of clinical management. An algorithm for the initial treatment of children with activity/attention problems, and symptoms refractory to methylphenidate. There is an appendix with recommended assessment instruments.

The TCMAP issued a revision of its 1998 recommendations for the medication treatment of children with ADHD with and without comorbid disorders.¹⁷ The TCMAP ADHD algorithm does not include psychosocial treatments or assessment of ADHD. Both the original and revised versions were based on a consensus conference of experts. The original version of this algorithm recommended a stimulant (methylphenidate or amphetamine) as the first stage of treatment. If one of these stimulants did not produce a satisfactory result, then stage 2 would involve the patient receiving the other stimulant not used in stage 1. Stage 3 was a trial of pemoline, and stage 4 was a trial of either bupropion or a tricyclic antidepressant. Stage 5 was the agent not used in stage 4, whereas stage 6 was treatment with an alpha agonist. Subsequently, an open trial of the feasibility of the algorithm for the treatment of primary ADHD resulted in the algorithm being well received by physicians, with good adherence through the first 2 stages but lower adherence in subsequent stages because of the small number of children progressing to this point.¹⁸ The major changes in the revised algorithm included elimination of pemoline as a treatment option, adding atomoxetine to the algorithm, and refining guidelines for treating ADHD with comorbid depression, aggressive behaviors, and tic disorders.

Magellan Health Services is a diversified specialty health care management organization whose behavioral health and employee assistance/work-life services cover more than 40 million individuals. The Magellan guidelines¹⁹ were written by Magellan to serve as an evidence-based framework for practitioners' clinical decision making with child, adolescent, and adult patients who have a diagnosis of ADHD. The 2010 revised guidelines were based on a review of the published literature, along with reported practitioner input. There is no mention of expert review. The recommendations are quite broad and are listed under the main areas of evaluation and treatment. Other guidelines, such as AAP and AACAP, are referenced.

Comparison of the Guidelines

In many respects the guidelines are quite different in their development, scope, and targets. These CPGs were

developed with different methodologies for different outcomes, and some were more focused on best cost outcomes. For example, different medications are available in Europe compared with the United States, and cultural differences would have influenced the development of guidelines. As mentioned previously, each of the guidelines involved one or more elements of proscribed CPG elements from among the Appraisal of Guidelines for

Research and Evaluation (AGREE) domains. The TCMAP algorithm, while admittedly not a CPF, is based only on limited expert consensus. The other CPGs include more or less the full range of the CPG procedures, including rigorous literature review and grading of the evidence and expert and stakeholder review, although these procedures are variable across the CPGs and involve different grading systems, review timeframes, and cross-disciplinary

Table 3. A Summary of Clinical Practice Guidelines for ADHD

1. Patient-Centered Care

- Treatment and care should consider an individual's needs and preferences and (in children) those of their parents or caregivers.
- All individuals with ADHD should be involved in decision making about their care and treatment in partnership with their health care professionals.
- Families and caregivers should be provided the information and support they need, and be encouraged to become involved in interventions.

2. Screening and Evaluation

- Screening for ADHD should be a part of every mental health evaluation and developmental screening for children and adolescents.
- Evaluation should consist of clinical interviews with the parent and patient, obtaining information about the patient's school or day-care functioning, assessment for comorbid psychiatric disorders, and review of the medical, social, and family history.
- For a diagnosis of ADHD, symptoms should meet the diagnostic criteria in DSM-IV or ICD-10 (hyperkinetic disorder) and be associated with at least moderate psychological, social and/or educational, or occupational impairment based on interviews, instruments, and/or direct observation in multiple settings.
- Severity should be measured and based on the number and severity of ADHD symptoms/behaviors, presence of psychiatric comorbidity, level of impairment, including pervasiveness (eg, occurrence in ≥ 2 important settings, including social, familial, educational, and/or occupational settings).
- The use of age-appropriate standardized instruments is suggested.
- If the patient's medical or educational history is unremarkable, laboratory testing, neurological testing, or psychological and neuropsychological tests are not mandatory for the diagnosis of ADHD.

3. Comprehensive Treatment Planning

- Patients, parents/caregivers, and educators should assist in the development of a comprehensive treatment program that addresses psychological, behavioral, and educational/occupational needs.
- Treatment selection (medication and/or psychosocial interventions) should consider severity, context of symptoms and impairment, patient age, preferences, capabilities, and personal circumstances of the patients, parents/caregivers.
- Medication trials should usually begin with stimulants (methylphenidate or amphetamine) and proceed to other stimulants and available FDA-approved medications (long-acting guanfacine or atomoxetine) in the case of non- or poor response or intolerable adverse events. Trials with off-label agents such as bupropion, alpha-adrenergic agonist, modafinil, or tricyclic antidepressants should be reserved for individuals who are unable to tolerate or respond to the FDA-approved agents listed previously.
- Cardiovascular evaluation prior to a stimulant trial should be based on the presence of medical history risk factors (eg, structural cardiac abnormalities, cardiovascular symptoms, family history of sudden death at an early age).
- Clinicians should regularly monitor medication response to treatment and titrate doses to control symptoms, reduce impairment and other behavioral targets, while minimizing adverse events.
- Clinicians should regularly monitor for adverse events, blood pressure, heart rate, weight, and height. Adolescents and adults should be monitored for illicit substance use. Both patients and families should be monitored for possible diversion of medication.
- Clinicians should treat comorbid psychiatric disorders or refer to specialists or other physicians for further evaluation and appropriate treatment.
- Psychosocial interventions should be considered for all patients with ADHD and their families, especially preschool children and those with coexisting problems with behavior, emotions, family, or occupation. For children, such interventions include parent management training and other operant behavioral interventions, and possibly social skills/problem-solving therapy and, for adults, cognitive behavioral therapies.
- Modification of educational placements and accommodations should be considered for all students with ADHD. Teachers should be trained to implement class or school-wide behavioral interventions and to assist in individual behavior management plans.

4. Follow-up

- Clinicians should follow-up with patients on medications regularly to determine if continued treatment is warranted and if effectiveness continues to be present.

review. For example, the British Psychopharmacology Guidelines include a very detailed indication of categories of evidence (for both causal relationships and treatment and observational relationships) and strength of recommendation ratings (based on the level of evidence above). Similarly, the Australian guidelines provide both categories of evidence and strength of recommendations. The AACAP guidelines do this as well but not as comprehensively. The AAP guidelines do neither explicitly, although we expect this to change with the upcoming revision. The NICE guidelines were the result of a guideline development group, which reviewed the evidence and developed the recommendations. The guidelines does not mention specific categories of evidence or strength of recommendations. The NICE guidelines do emphasize “person-centered care,” in which treatment and care take into account the needs and preferences of the patient and their family and involve them in decision making about their care. While there is little evidence that these considerations influence outcomes, concern about patient/family involvement is a general principle of medicine.²¹

The scope of CPGs differs. For example, some (AAP, AACAP, and TCMAP) are directed toward the evaluation and treatment of children and adolescents, while the NICE and Australian guidelines include adults in addition to younger populations. The British Psychopharmacology guidelines focus on older adolescents and adults. The guidelines including adults refer often to the more extensive literature and experience with children and adolescents.

Finally, the target consumer(s) appear to be differ among guidelines. The NICE guidelines appear to have the broadest targets, providing information for the patients, their families, and the providers (doctors and other mental health professionals, educators, and policy makers). The other CPGs for ADHD appear to target various combinations of physicians (pediatricians, other primary care physicians, and psychiatrists) and physicians. Because some CPGs were developed in the United Kingdom and Australia, and others in the United States, the availability of specific medications or stimulant formulations is different and may affect the recommendations. The cultural attitudes about medicating children or about the nature of ADHD may also differ among the countries where the CPG originated, with European countries being less predisposed to consider medication in children first.

Summary

Despite the differences between these CPGs for ADHD, they appear to be complementary and not inconsistent. Their parameters are summarized Table 3. Above all, the CPGs all

recommend a structured approach to diagnosis and treatment, attention to psychiatric comorbidity and other problems associated with ADHD across the lifespan, consideration of both medication (usually stimulants) and psychosocial therapies, and follow-up and monitoring of response. Finally, although less evidenced based than specific recommended therapies, a patient-centered approach respecting the preferences and decision-making capabilities of patients and their families in each of the tasks above is critical to acceptance of an ADHD diagnosis, adherence to treatment regimens, and optimal outcomes.

Conflict of Interest Statement

Oscar G. Bukstein, MD, MPH discloses no conflicts of interest.

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